The Golden Whisk Caper

Whodunnit? · Expressions & Equations · Teacher Guide and Answer Key

Unit	Time	Supplies	Group size
Expressions & Equations	20–40 minutes	Pencil, scrap paper, scissors	Teams of 2–3 (or solo)

Skills practiced: Solving multi-step equations, including distributing and variables on both sides.

General procedures

1. Give each student the **student sheet** (story and case board) and the **Show Your Work** page.
2. Print the **Clue Cards** (pages 3–7 of the student packet) and cut them apart on the dashed lines. Laminating them lets you reuse the set. Post them around the room.
3. Students visit the cards in any order. At each card they solve the problem on their work page, find their answer among the four choices, read the clue beside it, and cross things off the case board. **Every wrong choice leads to a false clue** that clears the real culprit, so a mistake shows up when the board does not narrow down to one suspect, one place and one method.
4. When a team thinks it knows, they bring you the three names. If they are wrong, let them hunt for the card where something went wrong.

Solution: Chef Dmitri took it, in the Pantry Tent, using this method: Hidden in a flour sack.

The reveal (read aloud): During the cake-cutting, **Chef Dmitri** ducked into the **Pantry Tent**, buried the Golden Whisk in a **flour sack**, and carried it out as a “delivery.” Every alibi and every method you cleared pointed straight at him. Sweet detective work!

Clue map: what each correct clue clears

Card	Clears
1	Suspect: Rosa Alvarez · Place: Oven Row
2	Method: Stashed in a picnic basket
3	Suspect: Mr. Okafor · Place: Judges’ Table
4	Method: Wheeled out in a mixer cart
5	Suspect: Ms. Lindqvist · Place: Tasting Tent
6	Method: Baked inside a giant pie
7	Suspect: Tess Brooks · Place: Frosting Station
8	Method: Posed as a cake topper
9	Suspect: Grandma June · Place: Cooler Truck
10	Method: Wrapped in a tablecloth

Answer Key

The Golden Whisk Caper · Correct choice and worked solution for each card

#	Problem	Correct choice	Worked solution	Clue it leads to
1	Solve: $3x - 7 = 8$	**D.** $x = 5$	$3x = 15$ $x = 5$	Rosa Alvarez was pulling scones out of the oven in Oven Row while three bakers watched.
2	Solve: $-2x + 5 = 11$	**C.** $x = -3$	$-2x = 6$ $x = -3$	Security checked every picnic basket at the gate. None held the Whisk.
3	Solve: $5(x - 1) = 2x + 7$	**B.** $x = 4$	$5x - 5 = 2x + 7$ $3x = 12$ $x = 4$	Mr. Okafor was scoring entries at the Judges' Table. His score sheets are time-stamped.
4	Solve: $7x + 4 = 2x - 16$	**D.** $x = -4$	$5x + 4 = -16$ $5x = -20$ $x = -4$	The mixer cart was weighed and wheeled back empty, so the Whisk never left in it.
5	Solve: $\frac{x + 4}{3} = 2$	**D.** $x = 2$	$x + 4 = 6$ $x = 2$	Ms. Lindqvist was sampling pies in the Tasting Tent. Her notebook records every bite.
6	Solve: $-3(x + 2) = 12$	**A.** $x = -6$	$x + 2 = -4$ $x = -6$	The giant pie was cut open on stage. Only blueberry filling inside.
7	Solve: $3x - 2(x - 4) = 14$	**A.** $x = 6$	$3x - 2x + 8 = 14$ $x + 8 = 14$ $x = 6$	Tess Brooks was piping frosting at the Frosting Station for a long line of kids.
8	Solve: $4x + 3 = 6x + 7$	**A.** $x = -2$	$3 = 2x + 7$ $-4 = 2x$ $x = -2$	The topper on the display cake is plastic. The Whisk was not disguised as one.
9	Twice a number plus 9 equals 5 times the number. Write and solve an equation to find the number, x.	**B.** $x = 3$	$2x + 9 = 5x$ $9 = 3x$ $x = 3$	Grandma June was handing out lemonade from the Cooler Truck all afternoon.
10	Solve: $8 - 3x = 23$	**C.** $x = -5$	$-3x = 15$ $x = -5$	Every tablecloth was shaken out and folded. Nothing was wrapped inside.

Wrong-Answer Map

The Golden Whisk Caper · What a wrong choice looks like, and the false clue it leads to

Card	Wrong choice	Likely error	False clue
1	A. $x = 15$	added 7 and 8 but forgot to divide	Chef Dmitri was checking his sourdough in Oven Row. Two bakers say he never left the oven door.
1	B. $x = -5$	sign error	Every flour sack was sliced open and sifted. No Whisk, just flour.
1	C. $x = \frac{1}{3}$	subtracted 7 instead of adding	Rosa Alvarez was counting sugar bags in the Pantry Tent. The inventory sheet has her handwriting.
2	A. $x = 3$	sign error dividing by -2	Chef Dmitri was munching a cookie in the Tasting Tent. A judge jotted down his comments on it.
2	B. $x = -8$	added 5 instead of subtracting	Mr. Okafor stopped by the Pantry Tent to find a stapler. The volunteer there remembers him.
2	D. $x = 8$	added 5 and sign error	The flour sacks are all still sealed with their factory stitching, so nothing was hidden inside.
3	A. $x = \frac{12}{7}$	added 2x instead of subtracting	A volunteer poured out the flour sack by the Pantry Tent. Not a single trophy was in it.
3	C. $x = \frac{8}{3}$	did not distribute to -1	Chef Dmitri was arguing about a score at the Judges' Table. The table monitor remembers every word.
3	D. $x = -4$	sign error	Ms. Lindqvist was interviewing the supply manager in the Pantry Tent. Her recorder was running.
4	A. $x = -\frac{4}{3}$	added 2x instead of subtracting	Tess Brooks was restocking cinnamon in the Pantry Tent all through the final round.
4	B. $x = 4$	sign error	Chef Dmitri was loading ice cream into the Cooler Truck. The driver's clipboard has his name.
4	C. $x = -\frac{12}{5}$	moved 4 with wrong sign	The flour delivery was weighed on arrival and again after the contest. The totals match exactly.
5	A. $x = 6$	forgot to subtract 4	Chef Dmitri was decorating cupcakes at the Frosting Station. Sprinkles were still stuck to his sleeve.
5	B. $x = -2$	sign error	A baker dumped the last flour sack into the mixing bowl and saw nothing but flour.
5	C. $x = 10$	added 4 instead of subtracting	Grandma June was borrowing eggs from the Pantry Tent, and the sign-out clipboard shows her name.
6	B. $x = 2$	sign error	A volunteer saw Rosa Alvarez grabbing vanilla in the Pantry Tent just as the timer buzzed.
6	C. $x = -2$	distributed -3 incorrectly	A food blogger filmed Chef Dmitri tasting jam in the Tasting Tent at the moment the Whisk disappeared.
6	D. $x = 6$	sign error in quotient	The flour sacks all still have their original tags, so none was opened and refilled.
7	B. $x = -6$	sign error	Mr. Okafor was sharpening pencils for the score sheets in the Pantry Tent. Shavings were still on the floor.
7	C. $x = 22$	distributed -2 incorrectly to -8	Chef Dmitri was timing his croissants at Oven Row. The oven timer is still set to his batch.
7	D. $x = 10$	did not multiply -4 by -2	The flour sacks were dusted for fingerprints. The only prints belong to the delivery driver.
8	B. $x = 5$	added 3 and 7 instead of subtracting	The cleanup crew emptied every flour sack into the compost bin and found only flour.

8	C. $x = \frac{2}{5}$	added 6x instead of subtracting	Tess Brooks was stacking paper cups in the Pantry Tent. A photo from 9:15 shows her there.
8	D. $x = 2$	sign error	Chef Dmitri was grabbing butter from the Cooler Truck. The cold-storage log shows his initials.
9	A. $x = \frac{9}{5}$	ignored 2x	A kid tossed a flour sack around during the pie-eating break. Nothing clunked inside.
9	C. $x = -3$	sign error	Chef Dmitri was waiting to talk to the judges at the Judges’ Table. Three bakers stood in line behind him.
9	D. $x = -\frac{9}{7}$	added 2x and 5x	Ms. Lindqvist ducked into the Pantry Tent to escape the sun. The tent guard wrote down the time.
10	A. $x = \frac{23}{5}$	combined 8 and $-3x$	Photos from the flour-sack race show every sack flat and floppy. None was hiding a trophy.
10	B. $x = 5$	sign error	Chef Dmitri was fixing a wobbly cake at the Frosting Station. A little girl helped him hold the plate.
10	D. $x = \frac{31}{3}$	added 8 instead of subtracting	Grandma June was hunting for her secret cinnamon in the Pantry Tent when the gasp went up.